Agentic AI: Technical Assignment

System Architecture & Automation Planning

Project: AI-Powered Islamic Digital Library
Assignment Duration: 1–2 Days
Purpose: Initial technical assessment and capability evaluation

1. Project Overview

We are developing an automated publishing and knowledge enrichment system for a digital library containing thousands of Islamic books in Arabic and other languages.
The objective is to process digitized books and automatically enrich them with structured metadata, detailed indexes, annotations, knowledge units and relationships to make their contents more discoverable and interconnected.
Scope clarification: OCR and initial text digitization are being handled by a separate team. This assignment focuses exclusively on post-OCR processing, AI automation, knowledge extraction and publishing workflows. The original book content must remain unchanged. All additional information will be maintained in a separate structured enrichment layer and linked to the corresponding book content.

2. Key Functional Requirements

A. Metadata Extraction

- Extract book title, author, editor, publisher, edition, publication year and other bibliographic information.
- Identify relevant information from prefaces, introductions, concluding pages and other sections.
- Identify missing or inconsistent metadata.

B. Book Family and Relationship Mapping

- Identify original/fundamental books and related works.
- Map relationships such as commentary (Sharh), supercommentary (Hashiyah), translation, summary and explanation.
- Establish hierarchical relationships and map corresponding sections or textual blocks across books.

C. Deep Indexing and Segmentation

- Identify books, volumes, chapters, subchapters and sections.
- Generate topic-based and semantic indexes.
- Segment content into paragraphs and individual knowledge units.
- Support different structural patterns across Islamic literature.

D. Arabic NLP and Search Enhancement

- Arabic normalization and diacritic handling.
- Morphological analysis, root identification and lemmatization.

  Entity recognition and disambiguation.

- Semantic search and contextual retrieval.
- Handling classical Arabic and variations in spelling and terminology.

E. Annotation Mapping

- Identify and map footnotes, endnotes, sidenotes and internal references.
- Link annotations to their corresponding text locations.

F. Genre-Specific Knowledge Extraction

Design dynamic extraction approaches for different types of Islamic literature:

Genre	Examples of knowledge units
Fiqh	Mas'alas, rulings, evidence, madhhab, scholarly opinions
Hadith	Hadith text, narrators, chains, grading, commentary
Tafseer	Quranic verses, explanations, interpretations
Seerah	Events, personalities, dates, locations
Aqeedah	Concepts, propositions, evidence
Islamic history	Events, personalities, chronology, places
Biographies	Scholars, teachers, students, birth/death details

Extracted information should retain references to the original book and precise source location wherever possible.

G. Knowledge Graph and Cross-Book Mapping

- Identify entities and relationships among books, scholars, concepts, events, rulings and references.
- Connect related knowledge units across different books.
- Support interconnected knowledge discovery.

3. Required Technical Assessment

Please propose an appropriate architecture and technology stack covering:

1. **System architecture:** Main components, modular design and scalability.
2. **Data flow:** From digitized book input through processing, enrichment, validation and publishing.
3. **Automation:** Python scripts, workflow orchestration, AI agents and task-specific pipelines.
4. **AI models:** Suitable commercial LLM APIs and open-weight models for the different processing tasks.
5. **Arabic NLP:** Relevant libraries, models and processing techniques.
6. **Knowledge infrastructure:** Relational databases, vector databases, graph databases and search engines.
7. **Multi-model orchestration:** Model comparison, independent validation, consensus, confidence scoring and discrepancy detection.
8. **Human review:** Mechanisms for reviewing and approving AI-generated outputs.
9. **Model adaptation:** Potential fine-tuning approaches, training data preparation and infrastructure

requirements for domain-specific open-weight models.

Please distinguish which tasks are best handled through conventional programming, NLP, LLMs or agentic workflows.

4. Expected Data Flow

Please illustrate a proposed workflow covering:

1. Receive digitized book text, PDF and associated identifiers.
2. Analyse document structure.
3. Extract and enrich metadata.
4. Identify book genre and related works.
5. Generate structural and semantic indexes.
6. Identify and map annotations and references.
7. Extract genre-specific knowledge units.
8. Identify entities and establish relationships.
9. Validate outputs using AI-based quality checks.
10. Send uncertain or sensitive results for human review.
11. Store approved information in the enrichment layer.
12. Make enriched books and extracted knowledge available for digital publishing and search.

5. Initial Deliverables (Within 1–2 Days)

This is a preliminary architecture and capability assessment, not a full system development assignment.
Please provide:

1. System Architecture Diagram

A high-level diagram showing the proposed system components and their interactions.

2. Automation and Data-Flow Diagram

A visual representation of the proposed processing pipeline, including AI agents, scripts, databases and human review.

3. Technology Recommendations

A concise list of recommended frameworks, tools, models, databases and their respective purposes.

4. Arabic NLP and AI Model Strategy (optional)

Recommendations for suitable models and tools, including commercial APIs and open-weight alternatives.

5. Knowledge Extraction Approach

A brief explanation of how you would implement genre-specific knowledge extraction, entity recognition and source-location mapping.

6. Model Training Approach

An initial proposal for preparing domain-specific training datasets and fine-tuning suitable open-weight models, including potential techniques such as LoRA/PEFT.

7. MVP / Proof of Concept (Optional)

If feasible, demonstrate one or two practical capabilities using the provided sample PDF and JSON, such as:

- Metadata extraction.

  Structural indexing.

- Knowledge-unit extraction.
- Entity and relationship identification.
- Source-location mapping.
- Structured JSON output.

A small prototype or technical demonstration is sufficient. Full implementation is not expected.

8. Initial Implementation Roadmap

Briefly outline development phases, major technical challenges, infrastructure requirements and indicative resource needs.

6. Technical Considerations

The proposed architecture should ensure:

- Original book content remains untouched.
- Enrichment data is stored separately and linked to the original content.
- Every extracted knowledge unit is traceable to its source.
- Original author statements are distinguished from AI-generated interpretations.
- Processing is modular, repeatable and scalable.
- AI-generated outputs support confidence assessment and human approval.

7. Reference Materials

The following materials will be shared with this assignment:

1. Sample Islamic book in PDF format.
2. Corresponding digitized book JSON file.

Please use these materials to understand the input structure and assess the feasibility of your proposed processing workflows.

8. Assessment Focus

The initial assessment will focus on:

- Understanding of the requirements.
- System architecture and technical problem-solving.
- Practical knowledge of AI automation and agentic workflows.
- Model selection and fine-tuning strategy.
- Appropriate technology choices and scalability considerations.
- Clarity and feasibility of the proposed implementation approach.

We welcome independent recommendations and alternative technical approaches where appropriate.